Reasoning

Mathematical Inequalities

Level-1

Q1 Directions:In the given questions, assuming the given statements to be true. Find which of the given two conclusions number I and II is/are definitely true and give your answer accordingly.

Statement: D < B ≤ # = F < 9 < 0

Conclusions:

I. # < 0

II. B ≤ F

(A) Only conclusion I follows

(B) Only conclusion II follows

(C) Both conclusions I and II follow

(D) Neither conclusion I nor II follows

(E) Either conclusion I or II follows

Q2 Directions:In the given questions, assuming the given statements to be true. Find which of the given two conclusions number I and II is/are definitely true and give your answer accordingly.

Statement: @ ≤ & ≤ 7 = % > 1 < #

Conclusions:

I. 7 ≥ @

II. 7 > 1

(A) Only conclusion I follows

(B) Only conclusion II follows

(C) Both conclusion I and II follow

(D) Neither conclusion I nor II follows

(E) Either conclusion I or II follows

Q3

Directions: Study the questions based on the information given below.

Statement: E < S ≤ ∞ = % < Y < @

Conclusions:

I. Y ≥ S

II. Y > E

(A) Only conclusion I follows

(B) Only conclusion II follows

(C) Both conclusion I and II follow

(D) Neither I nor II follows

(E) None of the above

Q4 Directions: Study the questions based on the information given below.

Statement: P = G < Q, E < L ≥ U, K ≤ G = R > L

Conclusions:

I. E < Q

II.G ≥ E

(A) only conclusion I follows

(B) only conclusion II follows

(C) Both conclusion I and II follow

(D) Neither conclusion I nor II follows

(E) either conclusion I or II follows

Q5 Directions: Study the questions based on the information given below.

Statement: X = B < N, B < O ≥ A, K ≤ O = M > B

Conclusions:

I. B < K

II. M > B

(A) Only conclusion I follows

(B) Only conclusion II follows

(C) Both conclusion I and II follow
(D) Neither conclusion I nor II follows
(E) Either conclusion I or II follows

Q6 Directions: Study the questions based on the information given below.
Statement: X = U < T, U < S ≥ I, K ≤ S = M > U
Conclusions:
I.U < K
II.M > K
(A) Only conclusion 1 follows
(B) Only conclusion 2 follows
(C) Either conclusion 1 nor 2 follows
(D) Neither conclusion 1 nor 2 follows
(E) If both conclusion I and conclusion II follows

Q7 **Directions:Analyze the data carefully and answer the questions accordingly.**
Statements: C < N < R > S, N< P
Conclusions:
I. P > S
II. R > C
(A) If only conclusion I follows
(B) If only conclusion II follows
(C) If either conclusion I or II follows
(D) If neither conclusion I nor II follows
(E) If both conclusions I and II follow

Q8 **Directions:Analyze the data carefully and answer the questions accordingly.**
Statements: F < P >Q = W, F ≤ Y, P > U
Conclusions:
I. Q > U
II. Y > W
(A) If only conclusion I follows
(B) If only conclusion II follows
(C) If either conclusion I or II follows
(D) If neither conclusion I nor II follows
(E) If both conclusions I and II follow

Q9

Directions:In the given questions, assuming the given statements to be true. Find which of the given two conclusions number I and II is/are definitely true and give your answer accordingly.

Statements: G > I = J > K = L ≤ M
Conclusions:
I. I > L
II. M ≥ K
(A) If only conclusion I follows
(B) If only conclusion II follows
(C) If either conclusion I or II follows
(D) If neither conclusion I nor II follows
(E) If both conclusions I and II follow

Q10 **Directions:In the following question assuming the given statements to be true, find which of the interpretation among given interpretation is/are definitely true and then give your reaction accordingly.**

Statements: M < N = O > P, Q > R < M, S ≤ P
Conclusions:
I. Q > P
II. N > S
III. R < O
IV. R ≥ O
(A) None is true
(B) Only IV is true
(C) Both II and III are true
(D) Either I or III and IV are true
(E) All are true

Q11 **Directions:In the following question assuming the given statements to be true, find which of the interpretation among given interpretation is/are definitely true and then give your reaction accordingly.**

Statements: $E \geq U = D; R < A < F; W \leq D; W > F$

Conclusions:

I. $U < R$

II. $E = W$

III. $E > W$

(A) Only II is True

(B) Only III is True

(C) Only I and II are True

(D) Either I or II is true

(E) Either II or III is true

Q12 **DIrections:In the following question assuming the given statements to be true, find which of the interpretation among given interpretation is/are definitely true and then give your reaction accordingly.**

Statements: $T \geq C \geq F; E = A < D; X > T; D < F = T$

Conclusions:

I. $F < E$

II. $C = F$

III. $A > T$

(A) Only I is True

(B) Only II is True

(C) Only III is True

(D) Only I and III are True

(E) None is true

Q13 **Directions:In the following question assuming the given statements to be true, find which of the interpretation among given interpretation is/are definitely true and then give your reaction accordingly.**

Statements: $Z > Y \geq X \geq K; K = L \geq M$; Which of the following are definitely true?

(A) $X > L$ (B) $Z > L$

(C) $K = Z$ (D) $K < Y$

(E) None of these

Q14 **Directions:In the following question assuming the given statements to be True, find which of the conclusion among given conclusions is/are definitely true and then give your answers accordingly.**

Statements: $P < Q \geq G; G \geq I \geq E; C \leq P; C > U$

Conclusions:

I. $U > I$

II. $P \leq E$

(A) Both I and II follow

(B) Either I or II follows

(C) Only I follows

(D) Only II follows

(E) Neither I nor II follow

Answer Key

Q1	(C)	Q8	(D)
Q2	(C)	Q9	(E)
Q3	(B)	Q10	(C)
Q4	(A)	Q11	(E)
Q5	(B)	Q12	(E)
Q6	(D)	Q13	(B)
Q7	(B)	Q14	(E)

Hints & Solutions

Q1 Text Solution:

Ans: C

Solution: Given statement: D < B ≤ # = F < 9 < 0

Conclusions: I. # < 0 (True), (As, # = F < 9 < 0, so the relationship between # and 0 is # < 0)

II. B ≤ F (True), (As, B ≤ # = F, so the relationship between B and F is B ≤ F)

Therefore, both conclusion 1 and 2 follow

Hence, option c is the correct answer.

Q2 Text Solution:

Ans: C

Solution: Given statement:@ ≤ & ≤ 7 = % > 1 < #

Conclusions: I. 7 ≥ @ (True), (As,@ ≤ & ≤ 7, so the relationship between 7 and @ is 7 ≥ @)

II. 7 > 1 (True), (As, 7 = % > 1, so the relationship between 7 and 1 is 7 > 1)

Therefore, both conclusion I and II follow

Q3 Text Solution:

Given statement: E< S ≤ ∞ = % < Y < @

Conclusions:

I. Y ≥ S (False), (As, S ≤ ∞ = % < Y, so the relationship between Y and S is Y > S).

II. Y > E (True), (As, E < S ≤ ∞ = % < Y, so the relationship between Y and E is Y > E)

Therefore, only conclusion II is true.

Hence, Option B is the correct answer.

Q4 Text Solution:

Ans: A

Solution: Given statement: P = G < Q, E < L ≥ U, K ≤ G = R > L

On combining the given statements, we get

K ≤ G = R > L ≥ U

I. **Conclusions: I.** E < Q (True), (As Q > G = R > L > E, so the relationship between E and Q is E < Q)

III. **II.** G ≥ E (False), (As G = R > L> E, so the relationship between G and E is G > E)

Therefore, only conclusion I is true.

Hence, option a is the correct answer.

Q5 Text Solution:

Ans: B

Solution: Given statement: X = B < N, B < O ≥ A, K ≤ O = M > B

On combining the given statements, we get:

K ≤ O = M > B < O ≥ A

Conclusions:

I. B < K (False), (As: K ≤ O = M > B < O ≥ A, so the relationship between B and K can't be determined)

II. M > B (True), (As: K ≤ O = M > B < O ≥ A, so the relationship between M and B is M > B)

Therefore, only conclusion II is true.

Hence, option b is the correct answer.

Q6 Text Solution:

Given statement: X = U < T, U < S ≥ I, K ≤ S = M > U

On combining the given statements, we get:

K ≤ S = M > U < S ≥ I

Conclusions:

I. U < K (False), (As: K ≤ S = M > U, so the relationship between U and K can't be determined)

I. **II.** M > K (False), (As: K ≤ S = M, so the relationship between M and K is M > K)

Therefore, Neither conclusion 1 nor 2 follows

Hence, Option D is the correct answer.

Q7 Text Solution:

I. P > N < R > S it is false

II. C < N < R it is true

so option B is correct

Q8 Text Solution:

I. U < P > Q False

II. Y ≥ F < P >Q = W False

so option D is correct

Q9 Text Solution:

I. I = J > K = L True

II. K = L ≤ M True

So the option E is correct

Q10 Text Solution:

Given Statements: M < N = O > P, Q > R < M, S ≤ P

On combining: M < N = O > P ≥ S, Q > R < M < N = O > P

Conclusions:

I. Q > P → False (as Q > R < M < N = O > P → Relation cannot be determined)

II. N > S → True (as N = O > P ≥ S→N>S)

III. R < O → True (as R < M < N = O → R < O)

IV. R ≥ O → False (as R < M < N = O → R < O)

Hence, the correct option is (C).

Q11 Text Solution:

Given statements: E ≥ U = D; R < A < F; W ≤ D; W > F On combining: E ≥ U = D ≥ W > F > A > R

Conclusions:

I. U < R → False (as U = D ≥ W > F > A > R → U ≥ W > R → U > R)

II. E = W → False (as E ≥ U = D ≥ W → E ≥ D ≥ W → E ≥ W)

III. E > W → False (as E ≥ U = D ≥ W → E ≥ D ≥ W → E ≥ W)

Since, conclusion II and III form complementary pair and E ≥ W.

Hence, the correct option is (E).

Q12 Text Solution:

Given statements: T ≥ C ≥ F; E = A < D; X > T; D < F = T

On combining: E = A < D < F ≤ C ≤ T < X; F = T

Conclusions:

I. F < E → False (as E = A < D < F → E < F)

II. C = F → False (as per the given information F = T & T ≥ C ≥ F)

III. A > T → False (as A < D < F ≤ C ≤ T → A < F ≤ C ≤ T → A < C ≤ T → A < T)

Hence, the correct option is (E).

Q13 Text Solution:

Given statements: Z > Y ≥ X ≥ K; K = L ≥ M; On combining: Z > Y ≥ X ≥ K = L ≥ M;

Conclusions:

I. X > L → False (as X ≥ K and K = L implies X ≥ L, thus a clear relation cannot be determined)

II. Z > L → True (as Z > Y ≥ X ≥ K; K = L implies Z > L)

III. K = Z → False (as Z > Y ≥ X ≥ K implies K < Z)

IV. K < Y → False (as Z > Y ≥ X ≥ K implies Y ≥ K, thus a clear relation cannot be determined)

Hence, the correct option is (B).

Q14 Text Solution:

Given statements: P < Q ≥ G; G ≥ I ≥ E; C ≤ P; C > U

On combining: U < C ≤ P < Q ≥ G ≥ I ≥ E

Conclusions:

I. U > I → False (as U < C ≤ P < Q ≥ G ≥ I → thus clear relation between U and I cannot be determined)

II. P ≤ E → False (as P < Q ≥ G ≥ I ≥ E → thus clear relation between P and E cannot be determined)

Hence, neither I nor II is True.